# Recovering the Pricing Kernel from Index Options:

## An Estimation Exercise on the Radon–Nikodym Derivative

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**Abstract**

Option prices identify the risk-neutral density $q$ of a future index level; historical returns identify the physical density $p$. Their ratio is, up to discounting, the projection of the Radon–Nikodym derivative $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ onto the terminal index level. This paper estimates that object from SPX options and quantifies how much of it the data actually determines.

Four results. First, the Breeden–Litzenberger estimator is consistent at order $h^2$ but ill-posed: we derive the noise amplification analytically and confirm on live quotes that naive second differencing of SPX mids yields a density with a noise-to-signal ratio of 47 and 39% negative mass. Second, an arbitrage-constrained SVI fit removes this entirely, producing a density integrating to 0.99997 whose mean matches the forward to $7 \times 10^{-5}$. Third, the estimated pricing kernel is non-monotone at the point estimate – the pricing-kernel puzzle – but its shape reverses within the 95% confidence interval of the equity premium, so *the puzzle is not identified at this sample size*. Fourth, we reconstruct the VIX from our own chain to within 0.07%, and show that the jump correction to the log-contract replication is 12.3% of implied variance, roughly 47% of the measured variance risk premium. Newey–West inference reduces the premium's $t$-statistic from 21.3 to 5.5.

**Keywords:** pricing kernel, risk-neutral density, Breeden–Litzenberger, SVI, variance risk premium, Radon–Nikodym derivative.

## 1 Introduction

Most empirical option work prices a contract under $\mathbb{Q}$. This paper does something different: it treats the change of measure itself as the object of estimation. The quantity of interest is

$$M(x) = e^{-rT}\,\frac{q(x)}{p(x)}, \tag{1}$$

the pricing kernel projected onto the terminal index level, where $q$ is extracted from a cross-section of option prices and $p$ from a time series of returns. Theory predicts $M$ should decrease in wealth. A substantial empirical literature – Jackwerth [2000], Aït-Sahalia and Lo [2000], Rosenberg and Engle [2002] – finds it does not, a finding known as the pricing kernel puzzle.

Our contribution is less a new estimate than a careful accounting of what the data supports. Three methodological points organise the paper.

**The extraction is an ill-posed inverse problem, and we measure how badly.** Section 2 derives the noise amplification of the Breeden–Litzenberger second difference in closed form; Section 3 validates it against known densities; Section 5 confirms the prediction on live SPX quotes. The naive estimator is not merely noisy – on a five-point strike grid the noise exceeds the signal by a factor of 47.

*Code and data-processing scripts: `https://github.com/aryanbhujbal24/pricing-kernel`. All results in this paper are reproduced by running the five numbered scripts in order.

**The physical density must be conditional.** A kernel density on historical horizon returns is *unconditional*: it describes an average day since 1990, when volatility averaged 18%. Our $q$ is conditional on a day when implied volatility was 13.5%. Dividing one by the other measures the gap between today and an average day. We report both estimators and treat their difference as a result.

**The headline finding is a negative one.** The estimated kernel is non-monotone. But the physical drift enters directly, and the equity premium is estimated with a standard error of 2.97% per year. Sweeping the drift across its own confidence interval flips the kernel from strongly non-monotone to essentially monotone decreasing. We therefore cannot reject monotonicity, and we argue that a point estimate reported without this sensitivity claims a discovery the data has no power to make.

Section 2 derives the required results. Section 3 validates the estimator on synthetic data. Section 4 describes the data. Sections 5–7 present the surface, the kernel, and the variance risk premium. Section 8 states the limitations; Section 9 concludes.

# 2 Theory

## 2.1 Setup and the projected kernel

Fix a horizon $T$ and a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$. Let $S_t$ be an index paying continuous dividend yield $q_d$, let $r$ be constant, and write $F = S_0 e^{(r-q_d)T}$ for the forward. By the fundamental theorem of asset pricing there exists $\mathbb{Q} \sim \mathbb{P}$ under which discounted gains are martingales; write $\xi = \mathrm{d}\mathbb{Q}/\,\mathrm{d}\mathbb{P}|_{\mathcal{F}_T}$ and $M_T = e^{-rT}\xi$, so that $\mathrm{price}_0(X) = e^{-rT}\mathbb{E}^{\mathbb{Q}}[X] = \mathbb{E}^{\mathbb{P}}[M_T X]$.

Under Black–Scholes dynamics, Girsanov's theorem gives $\xi$ explicitly as a decreasing power function of $S_T$ with exponent set by the market price of risk $\lambda = (\mu - r + q_d)/\sigma$. In general $\xi$ need not be a function of $S_T$, and option prices cannot identify it if it is not. What they identify is its projection.

**Proposition 2.1** (Projection)**.** *Let $p, q$ be the densities of $S_T$ under $\mathbb{P}, \mathbb{Q}$ with $p > 0$ on the support. Then $\mathbb{E}^{\mathbb{P}}[\xi \mid S_T = x] = q(x)/p(x)$, and hence*

$$M(x) := \mathbb{E}^{\mathbb{P}}[M_T \mid S_T = x] = e^{-rT}\frac{q(x)}{p(x)}. \tag{2}$$

*Proof.* For bounded measurable $g$, changing measure and then conditioning,

$$\int g\, q = \mathbb{E}^{\mathbb{Q}}[g(S_T)] = \mathbb{E}^{\mathbb{P}}[\xi\, g(S_T)] = \mathbb{E}^{\mathbb{P}}\big[\mathbb{E}^{\mathbb{P}}[\xi \mid S_T]\, g(S_T)\big] = \int \mathbb{E}^{\mathbb{P}}[\xi \mid S_T = x]\, g(x)\, p(x)\, \mathrm{d}x.$$

Equality for all such $g$ forces $q(x) = \mathbb{E}^{\mathbb{P}}[\xi \mid S_T = x]\, p(x)$ a.e. □

Proposition 2.1 licenses the exercise and simultaneously bounds it: any component of $\xi$ orthogonal to $\sigma(S_T)$ – a separate volatility risk premium, for instance – is integrated out and invisible. A non-monotone estimate of $M$ is therefore not automatically evidence against theory; it may be evidence that $S_T$ is not a sufficient statistic.

## 2.2 Breeden–Litzenberger and its error structure

**Theorem 2.2** (Breeden and Litzenberger, 1978)**.** *If $S_T$ admits a $\mathbb{Q}$-density $q$ with $\mathbb{E}^{\mathbb{Q}}[S_T] < \infty$, then*

$$\frac{\partial C}{\partial K} = -e^{-rT}\mathbb{Q}(S_T > K), \qquad \frac{\partial^2 C}{\partial K^2} = e^{-rT} q(K). \tag{3}$$

*Proof.* $C(K) = e^{-rT}\int_K^\infty (x-K)q(x)\,\mathrm{d}x$. The integrand vanishes at $x = K$, so differentiating under the integral leaves only the explicit $K$-dependence, giving the first identity; differentiating again produces the boundary term $e^{-rT}q(K)$. Dominated convergence applies since the difference quotients of $(x-K)^+$ are bounded by 1 and $q$ is integrable. □

Financially, the finite-difference version is a butterfly spread: an infinitesimally tight butterfly is an Arrow–Debreu security and the market quotes its price. Non-negativity of $q$ requires $C$ convex in $K$; this is the butterfly no-arbitrage condition, and quoted chains violate it routinely.

Define the estimator on an equally spaced grid,

$$\hat{q}(K) = e^{rT}\,\frac{C(K+h) - 2C(K) + C(K-h)}{h^2}. \tag{4}$$

**Proposition 2.3** (Truncation)**.** *With exact prices and $C \in \mathcal{C}^4$, $\hat{q}(K) = q(K) + \frac{h^2}{12}q''(K) + O(h^4)$.*

*Proof.* Odd terms cancel in $C(K+h) - 2C(K) + C(K-h) = h^2C'' + \frac{h^4}{12}C'''' + O(h^6)$; apply $e^{rT}C'' = q$ and $e^{rT}C'''' = q''$. □

**Proposition 2.4** (Noise amplification)**.** *If observed prices carry i.i.d. errors of variance $\varepsilon^2$, then*

$$\mathrm{Var}[\hat{q}(K)] = \frac{6e^{2rT}\varepsilon^2}{h^4}, \qquad \mathbb{E}\int |\hat{q} - q|\,\mathrm{d}K \approx e^{rT}\sqrt{\tfrac{12}{\pi}}\,\frac{\varepsilon}{h^2}(K_{\max} - K_{\min}). \tag{5}$$

*Proof.* The second difference carries noise of variance $(1+4+1)\varepsilon^2$; divide by $h^2$ and multiply by $e^{rT}$. For the integrated error use $\mathbb{E}|N(0,s^2)| = s\sqrt{2/\pi}$ with $s$ the resulting standard deviation. □

Total error behaves as $ah^2 + b\varepsilon h^{-2}$, minimised at an interior $h^\star \propto \varepsilon^{1/4}$ that *widens* as quotes get noisier. The correct response to noise is therefore not a finer grid but regularisation: fit a smooth arbitrage-free surface in implied-volatility space, where the fitted function is flat and well-scaled, and differentiate it analytically.

### 2.3 Static replication and the log contract

**Theorem 2.5** (Carr and Madan, 2001)**.** *For $f \in \mathcal{C}^2$ and any $\kappa > 0$,*

$$f(x) = f(\kappa) + f'(\kappa)(x-\kappa) + \int_0^\kappa f''(K)(K-x)^+\,\mathrm{d}K + \int_\kappa^\infty f''(K)(x-K)^+\,\mathrm{d}K. \tag{6}$$

*Proof.* Taylor with integral remainder. For $x > \kappa$, $f(x) = f(\kappa) + f'(\kappa)(x-\kappa) + \int_\kappa^x (x-K)f''(K)\,\mathrm{d}K$, and $(x-K)^+$ vanishes for $K > x$; symmetrically for $x < \kappa$. □

Applying (6) to $f(x) = \log(x/F)$, for which $f''(K) = -1/K^2$, with $\kappa = F$ so the linear term has zero $\mathbb{Q}$-expectation:

$$-\mathbb{E}^{\mathbb{Q}}\left[\log\frac{S_T}{F}\right] = e^{rT}\int_0^\infty \frac{O(K)}{K^2}\,\mathrm{d}K, \tag{7}$$

with $O(K)$ the out-of-the-money price. The $1/K^2$ weight is not a convention; it is $|f''|$ for the logarithm. For a continuous semimartingale, Itô gives $\mathrm{d}\log S_t = \mathrm{d}S_t/S_t - \frac{1}{2}\sigma_t^2\,\mathrm{d}t$, so

$$\mathbb{E}^{\mathbb{Q}}\left[\int_0^T \sigma_t^2\,\mathrm{d}t\right] = -2\mathbb{E}^{\mathbb{Q}}\left[\log\frac{S_T}{F}\right] = 2e^{rT}\int_0^\infty \frac{O(K)}{K^2}\,\mathrm{d}K. \tag{8}$$

**The VIX correction term.** Strikes are discrete and $F$ is not one of them. Let $K_0$ be the largest listed strike below $F$ and re-expand there. Splitting the logarithm and applying Theorem 2.5 at $\kappa = K_0$ – where the linear term now contributes $-2(F/K_0 - 1)$ – and writing $u = F/K_0 - 1$ so that $-2\log(K_0/F) = 2\log(1+u) = 2u - u^2 + O(u^3)$, the $\pm 2u$ terms cancel and

$$\sigma^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2} e^{rT} O(K_i) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2, \tag{9}$$

which is the published CBOE formula [CBOE, 2019]. The much-copied correction term is simply the second-order cost of expanding around a listed strike rather than the forward.

### 2.4 What jumps do

Equation (8) assumed continuity. For a semimartingale with relative jumps $J$, Itô's lemma gains a term $\sum_t [\log(1+J_t) - J_t]$, and (8) becomes

$$\underbrace{2e^{rT}\int \frac{O(K)}{K^2}\,\mathrm{d}K}_{\text{measured}} = \underbrace{\mathbb{E}^{\mathbb{Q}}\Big[\int_0^T \sigma_t^2\,\mathrm{d}t + \sum_t J_t^2\Big]}_{\mathbb{E}[\mathrm{QV}]} - \frac{2}{3}\mathbb{E}^{\mathbb{Q}}\Big[\sum_t J_t^3\Big] + O(J^4), \tag{10}$$

using $\log(1+J) - J = -\frac{1}{2}J^2 + \frac{1}{3}J^3 - \cdots$. Three consequences. The log-contract price is not expected quadratic variation. Index jumps skew negative, so $\mathbb{E}^{\mathbb{Q}}[\sum J^3] < 0$ and the correction is *positive*: implied variance systematically overstates $\mathbb{E}[\mathrm{QV}]$. And since the variance risk premium is measured as implied minus realised, part of any measured premium is this jump term rather than compensation for variance risk. Section 7 quantifies it.

## 3 Validating the estimator on synthetic data

An estimator untested against a known answer is not an estimator. We therefore validate (4) against two densities we know exactly before touching market data.

**Black–Scholes: correctness.** With $S_0 = 5000$, $T = 0.5$, $r = 4\%$, $q_d = 1.5\%$, $\sigma = 20\%$ and strikes spanning $\pm 4$ standard deviations, the extractor recovers the lognormal with empirical convergence order $1.999 \pm 0.001$ across four decades of $h$; at $h = 1$ the maximum pointwise error is $1.0 \times 10^{-10}$. Proposition 2.3 predicts not only the order but the constant: the measured leading coefficient matches $q''(K)/12$ to within $4 \times 10^{-5}$ relative at $h = 25$. An estimator can be second-order accurate for the wrong reason; matching the constant rules that out.

**Black–Scholes with noise: ill-posedness.** Perturbing prices by i.i.d. noise of size $\varepsilon$ and averaging over 200 replications, the measured integrated error matches (5) to three significant figures wherever noise dominates (at $\varepsilon = 0.05$, $h = 10$: predicted 5.982, measured 5.926). Total error is U-shaped in $h$ with the optimum at $h^\star = 100, 200, 400$ for $\varepsilon = 0.005, 0.05, 0.25$, consistent with the $\varepsilon^{1/4}$ scaling. At five cents of noise on a ten-point grid, 44% of the extracted density is negative.

**Heston: a non-lognormal truth.** Black–Scholes is a forgiving test because it has no smile. We therefore repeat the exercise under Heston ($v_0 = 0.04$, $\kappa = 2$, $\theta = 0.045$, $\sigma_v = 0.55$, $\rho = -0.75$; Feller deliberately violated, as real calibrations usually are). Prices come from Gil-Pelaez inversion of the characteristic function in the Albrecher et al. [2007] formulation; the true density comes from inverting the same characteristic function directly. Recovery on a 25-point grid gives $L^1$ error $2.0 \times 10^{-4}$ and $L^\infty$ error $2.6 \times 10^{-7}$, with the negative skew

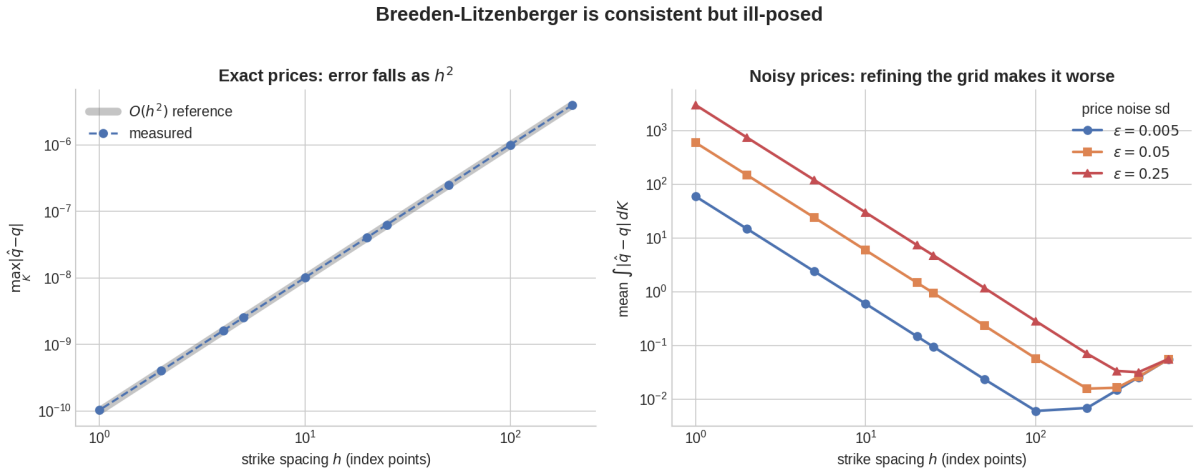



**Figure 1:** Left: on exact prices the error falls as $O(h^2)$, the grey reference lying underneath the measured points. Right: with quote noise the error is U-shaped in $h$, and refining the grid past $h^\star$ makes matters worse. The two panels together are the argument for regularisation.

correctly reproduced. As a further check, Heston collapses onto Black–Scholes as $\sigma_v \to 0$ to within $6.7 \times 10^{-7}$ in price.

# 4 Data

A single CBOE delayed-quote snapshot of SPX and SPXW options, 29,858 contracts across 57 expiries, with quotes as of the close on 25 September 2026 and the index at 7743.41. We use SPX monthlies, which are European and AM-settled – Breeden–Litzenberger assumes European exercise, so SPY, being American, is unusable for this purpose. Daily S&P 500 closes from 1927 and daily VIX closes from 1990 supply the time-series half.

Rather than importing an external discount curve we recover the forward and discount factor from the quotes themselves, regressing the parity spread $C - P$ on $K$ over strikes within 5% of spot, weighted by inverse quote width. This is self-consistent: it returns the discount rate the option market is actually using.

| Expiry | $T$ | strikes | $F$ | $D$ | implied $r$ | implied $q_d$ | parity RMSE |
|---|---|---|---|---|---|---|---|
| 2026-10-16 | 0.0575 | 482 | 7756.6 | 0.99754 | 4.29% | 1.33% | 0.078 |
| 2026-11-20 | 0.1534 | 458 | 7782.6 | 0.99306 | 4.54% | 1.25% | 0.147 |
| 2026-12-18 | 0.2301 | 455 | 7803.3 | 0.98965 | 4.52% | 1.17% | 0.108 |

**Table 1:** Put-call parity diagnostics. Implied rates and dividend yields are outputs, not inputs; the dividend yields near 1.2% are correct for SPX. Parity residuals of 0.1 index points on a 7800 index are about 1.3 basis points.

All subsequent cross-sectional work uses the December 2026 expiry, the longest monthly and therefore the widest in strikes and the best behaved for kernel estimation.

# 5 The arbitrage-free surface and the risk-neutral density

## 5.1 The naive estimator fails as predicted

Applying (4) directly to December mid quotes on the five-point strike grid, with a median half-spread of 0.65: Proposition 2.4 predicts a noise standard deviation of $3.7 \times 10^{-2}$ in $\hat{q}$, against

a peak density of $7.9 \times 10^{-4}$ implied by the 13.5% ATM volatility. The noise-to-signal ratio is 47, and 39% of the extracted density is negative (Figure 2). Raw mids also violate convexity at 35% of adjacent strike triples.

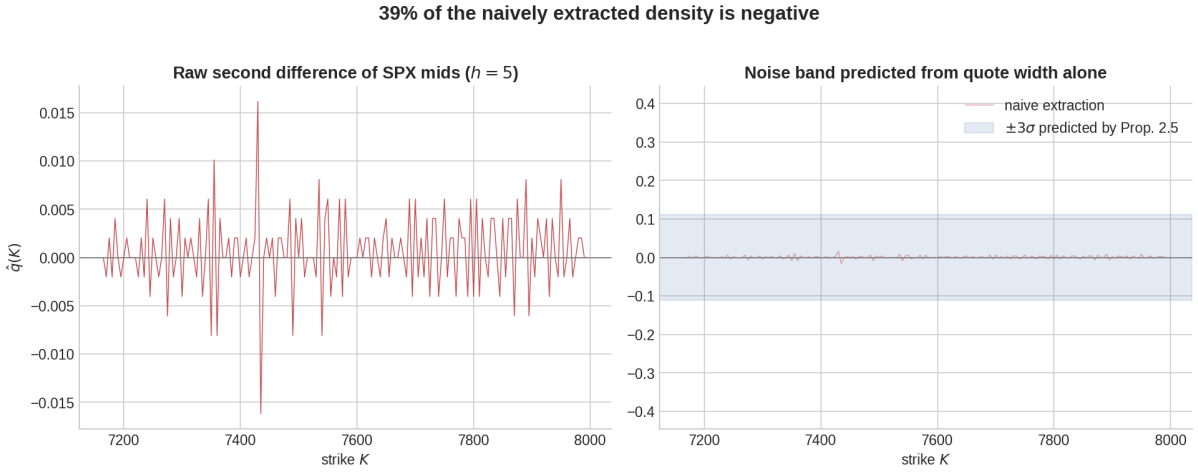



**Figure 2:** Naive second differencing of SPX December mids. The shaded band is the $\pm 3\sigma$ region predicted by Proposition 2.4 from the quote width alone, with no fitting. The estimator is not noisy; it is noise.

The measured 39% negative fraction falls short of the 49% predicted under i.i.d. errors. Inverting the measured fraction implies an effective independent noise of 0.028 index points against 0.375 implied by the spread. Quote errors are therefore strongly positively correlated across adjacent strikes – consistent with market makers quoting from a common fitted smile – and the second difference cancels most of the common component. The i.i.d. model is conservative here, not optimistic.

## 5.2 Constrained SVI

We fit raw SVI [Gatheral, 2004] in total variance, $w(k) = a + b[\rho(k-m) + \sqrt{(k-m)^2 + s^2}]$ with $k = \log(K/F)$, under hard constraints: $b \geq 0$, $s > 0$, $w > 0$, Lee's wing bound $b(1 + |\rho|) \leq 2$ [Lee, 2004], and the Gatheral and Jacquier [2014] butterfly condition $g(k) \geq 0$ on a dense grid extending 25% beyond the data. We parameterise $\rho$ as $\tanh(\cdot)$ so $|\rho| < 1$ holds structurally, and use sixteen random starts since the objective is not convex. Observations are weighted by inverse variance, propagating the quote half-width into total-variance units via $\mathrm{d}w = 2\sigma T \, \mathrm{d}C/\mathrm{vega}$.

The density is then read off analytically,

$$q(K) = \frac{g(k)}{K\sqrt{2\pi w(k)}} \exp\!\left(-\tfrac{1}{2} d_2^2\right), \qquad d_2 = -\frac{k}{\sqrt{w}} - \frac{\sqrt{w}}{2}, \tag{11}$$

verified against finite differences of the fitted slice's own call prices to $1.7 \times 10^{-6}$ relative. No market data is differentiated at any point.

**Quote reliability filter.** We drop 29 of 455 quotes priced below \$1 or with relative spread above 15%. These extend to $k = -1.72$, roughly 26 standard deviations out at this maturity, trading near fifty cents with spreads exceeding 30% of price; their implied volatilities are a tick-size artefact. This is a stated modelling choice, not data cleaning. The fitted range is $k \in [-0.774, +0.132]$.

Fitted parameters are $a = -0.0227$, $b = 0.1052$, $\rho = -0.0830$, $m = 0.0520$, $s = 0.2459$. The resulting density has total mass 0.99997 and a mean within 0.56 index points of the forward ($7 \times 10^{-5}$ relative), with skewness $-1.99$ and excess kurtosis 16.0. Neither the mass nor the mean

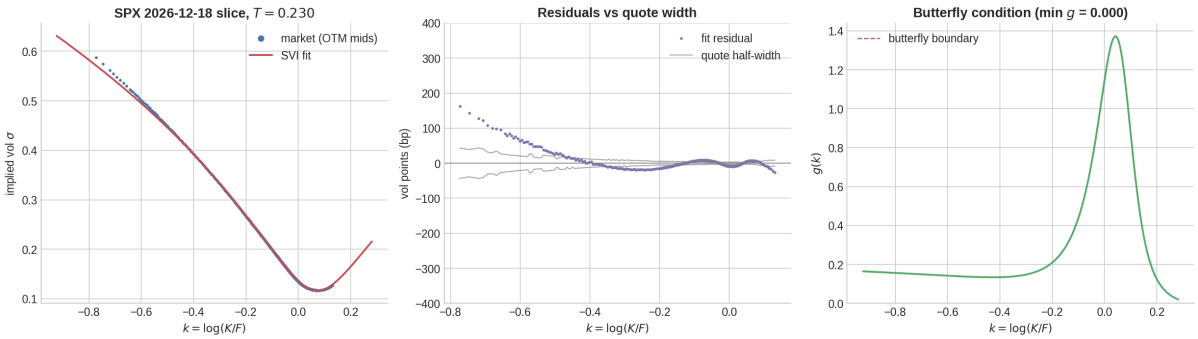



**Figure 3:** Fitted slice, residuals against the quote half-width, and the butterfly function $g$. The fit sits inside the quote band from $k \approx -0.4$ upward; the error is concentrated in the deep left wing.

| Region | $n$ | RMSE (bp) | inside bid-ask |
|---|---|---|---|
| core, $\lvert k\rvert \le 0.15$ | 288 | 6.6 | 26.7% |
| left, $-0.5 \le k < -0.15$ | 104 | 13.5 | 22.1% |
| deep left, $k < -0.5$ | 34 | 77.7 | 0.0% |
| all | 426 | 23.6 | 23.5% |

**Table 2:** Fit error by moneyness. The aggregate figure is misleading alone: in the core the fit is 6.6 bp, and essentially all the error lives in the deep left wing, where raw SVI's asymptotically *linear* total variance cannot match the observed *concave* wing. The low inside-spread fractions reflect a median SPX quote half-width of only 3.9 bp.

is imposed by the fit, so both are genuine checks: a five-parameter family is under no obligation to integrate correctly.

# 6 The physical density and the pricing kernel

## 6.1 Two estimators of $p$

The horizon is 59 NYSE sessions, $T = 0.2301$. We estimate $p$ two ways.

The *naive* estimator is a Gaussian kernel density on overlapping 59-day log returns since 1990. It is unconditional. Its nominal $n = 9{,}192$ is 156 independent observations in disguise, since 59-day returns sampled daily overlap by 58 of 59 days.

The *filtered* estimator follows Barone-Adesi et al. [2008]: fit GARCH(1,1) with Student-$t$ innovations to daily returns, then bootstrap standardised residuals forward 59 days from *today's* conditional variance. The fit gives $\alpha = 0.102$, $\beta = 0.892$, persistence 0.994, $\nu = 6.15$, with an unconditional volatility of 23.4% against a current conditional volatility of 11.6%; the simulated horizon distribution mean-reverts upward and lands at 13.8% annualised, with skewness $-0.81$ and kurtosis 9.4.

*Remark* 6.1 (Centring the residual pool). Standardised residuals do not have mean zero in sample, because high-variance days carry systematically signed residuals. Bootstrapping them raw injects an uncontrolled drift on top of the mean parameter. Centring the pool makes the horizon drift exactly $\mu h$, so the drift becomes an explicit assumption rather than an artefact of the filter. This was detected because the base case and a nominally identical sensitivity scenario disagreed.

The premium is visible directly in tail probabilities: $\mathbb{Q}(S_T < 7000) = 6.20\%$ against $\mathbb{P} = 3.96\%$ (ratio 1.56), while $\mathbb{Q}(S_T > 8700) = 2.02\%$ against $\mathbb{P} = 5.41\%$ (ratio 0.37). That asymmetry is the kernel.

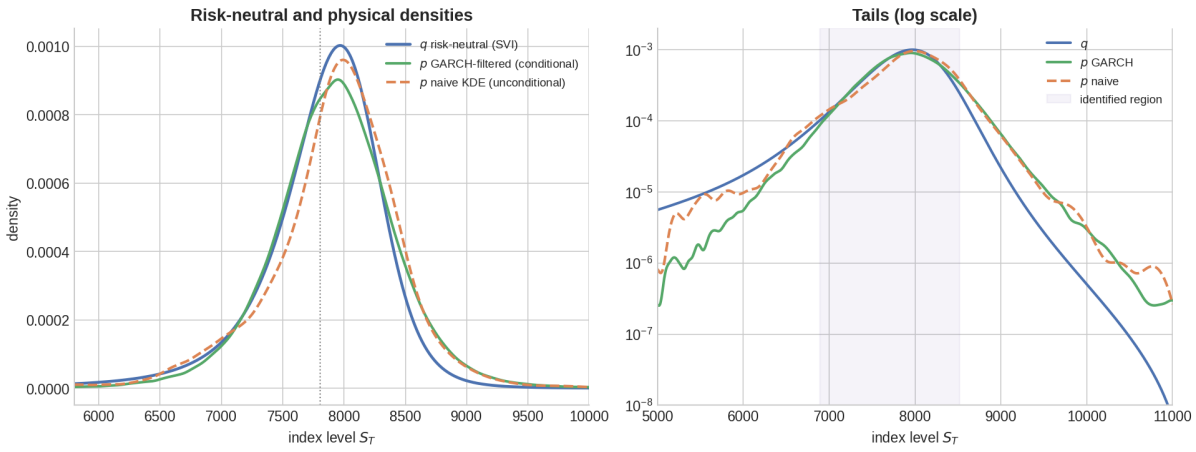



**Figure 4:** Risk-neutral and physical densities. $q$ has the fatter left tail and the thinner right tail: the market prices drawdowns above their historical frequency and rallies below it.

### 6.2 The kernel, and why its shape is not identified

On the identified region – the central 90% of $q$, $[6893, 8518]$ – the estimated kernel (2) is non-monotone: increasing over 41.3% of the region with the filtered $p$ and 39.8% with the naive one. Taken at face value this is the pricing kernel puzzle.

It should not be taken at face value. The physical drift enters $p$ directly and is the least well determined input in the exercise: the 1990–2026 sample mean is 8.41% per year with a standard error of 2.97%, a 95% interval of $[2.59\%, 14.23\%]$.

| Drift scenario | $\mu$ (annual) | fraction of region increasing |
|---|---|---|
| $\hat{\mu} - 2\,\text{s.e.}$ | 2.47% | 51.8% |
| risk-neutral drift | 3.35% | 50.2% |
| $\hat{\mu}$ | 8.41% | 39.9% |
| $\hat{\mu} + 2\,\text{s.e.}$ | 14.35% | **2.7%** |
| GARCH constant mean | 18.93% | 2.2% |

**Table 3:** Sensitivity of kernel monotonicity to the assumed physical drift. Within the 95% interval of the equity premium the verdict reverses completely.

At the top of the interval the kernel is essentially monotone decreasing, exactly as theory predicts; at the bottom it is strongly non-monotone. We therefore cannot reject monotonicity. The bootstrap band in Figure 5 reaches the same conclusion by a different route: it is wide enough to contain a monotone decreasing kernel across most of the region.

One trap deserves comment. The GARCH constant-mean estimate is 18.93% per year against an 8.41% sample mean, because GARCH weights low-variance days most heavily and low-variance days are bull markets. It is biased upward as an equity premium estimate. It also sits ready-made in the fitted object, and using it – as a naive pipeline would – produces a cleanly monotone kernel for entirely the wrong reason.

## 7 Model-free implied variance and the variance risk premium

### 7.1 Reconstructing the VIX

The snapshot's SPXW expiries at 28 and 35 days bracket 30, so (9) can be run on our own chain and checked against the published index. Following the CBOE recipe exactly – forward from the strike with the smallest $|C - P|$, $K_0$ the largest strike below it, OTM options subject to the

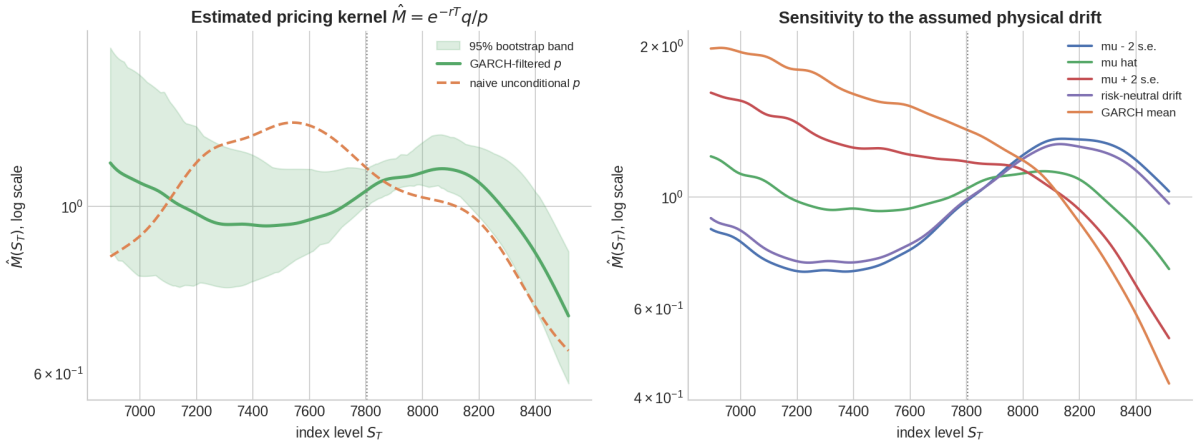



**Figure 5:** Left: estimated kernel with a 95% bootstrap band over 120 resamples of the residual pool. Right: the same kernel under drift assumptions spanning the equity premium's confidence interval. The shape is set by the drift, not by the option data.

two-consecutive-zero-bid stopping rule, and interpolation in total variance – we obtain

$$\text{VIX}_{\text{reconstructed}} = 14.880 \qquad \text{against} \qquad \text{VIX}_{\text{published}} = 14.87,$$

an error of +0.07%. This validates the derivation and the implementation together, and licenses using the published VIX history as a model-free implied variance series – which is what makes a 36-year time series possible from a single option snapshot.

**Truncation, quantified.** The $1/K^2$ weight loads on deep OTM puts, exactly where chains run out. The share of the variance sum by moneyness is 4.6% for $K/F < 0.70$, 9.1% for 0.70–0.85, 19.8% for 0.85–0.95, 64.3% for 0.95–1.05 and 2.1% above. So 4.6% of the answer comes from the region most exposed to truncation.

### 7.2 The jump correction

Refitting the 35-day slice through the Section 5 pipeline, the risk-neutral log return has variance 0.00242 and third central moment $-4.46 \times 10^{-4}$ (skewness $-3.74$). Applying (10):

| | |
|---|---|
| log-contract implied volatility | 15.90% |
| jump-corrected $\sqrt{\mathbb{E}[\text{QV}]}$ | 14.89% |
| correction as share of implied variance | 12.3% |

Scaled to the sample mean implied variance, the correction is 0.00537 in variance units against a measured mean premium of 0.01136 – roughly 47% of the measured premium is the jump term rather than compensation for variance risk. Two caveats travel with that figure: it is extrapolated from a single cross-section, and it is an *upper bound*, since attributing the entire third moment to jumps ignores that a leverage-driven stochastic-volatility diffusion generates negative skew without any jumps at all.

### 7.3 Testing the premium

We form $\text{VRP}_t = \text{IV}_t^2 - \text{RV}_t^2$ with IV the VIX and RV the subsequently realised variance over 21 sessions, giving 9,226 daily observations from 1990 to 2026. Mean implied volatility is 20.9% against mean subsequently realised volatility of 18.0%.

The premium survives, but a paper reporting 21.3 would be reporting an artefact of overlap.

| Specification | lags | mean | s.e. | $t$ |
|---|---|---|---|---|
| Naive, no HAC | 0 | 0.01136 | 0.00053 | **21.30** |
| Newey–West, lag = window | 21 | 0.01136 | 0.00206 | 5.52 |
| Newey–West, lag = 2× window | 42 | 0.01136 | 0.00232 | 4.90 |
| Newey–West, Andrews plug-in | 196 | 0.01136 | 0.00228 | 4.99 |

**Table 4:** Mean variance risk premium and standard errors. Overlapping windows make the series autocorrelated by construction even under the null; HAC inflates the standard error 3.9× and the $t$-statistic falls from 21.3 to about 5. Lag selection follows Newey and West [1987, 1994].

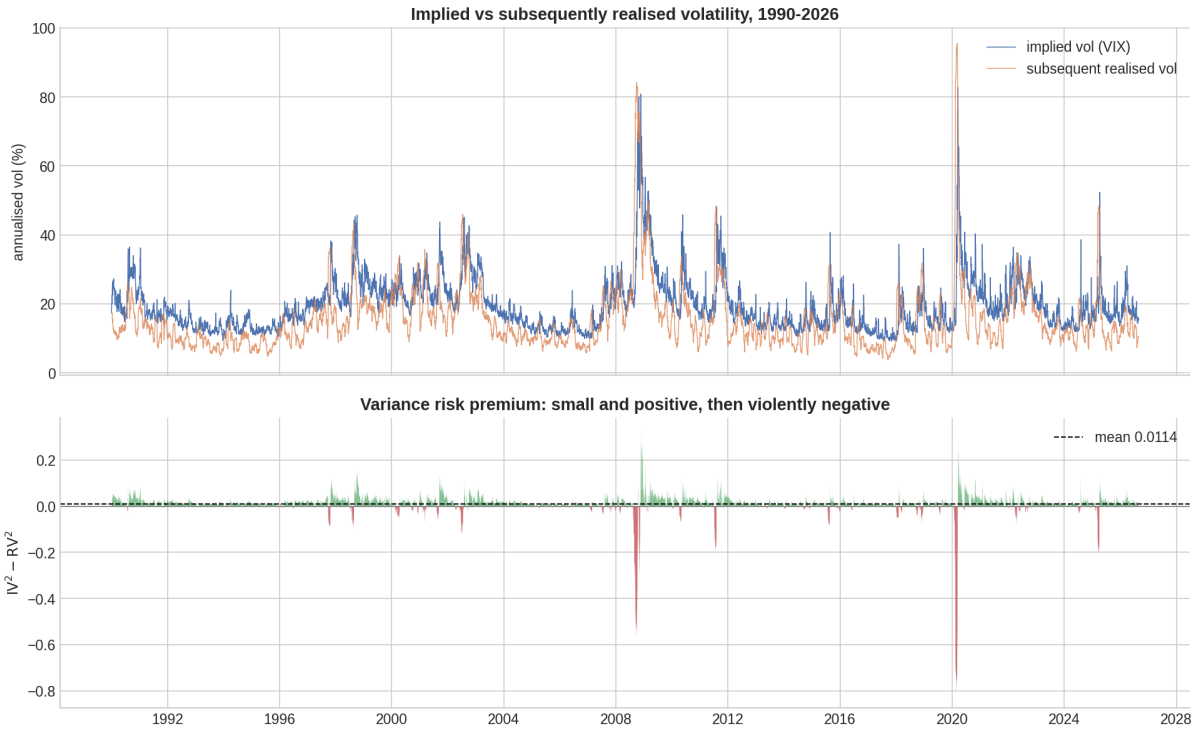



**Figure 6:** Implied versus subsequently realised volatility, and the resulting premium. Positive most of the time; violently negative in 2008 and 2020.

### 7.4 Shape, and why the mean is a weak summary

The premium is positive on 85.9% of days, with skewness −8.38 and excess kurtosis 109. The worst observation, −0.816, is 26 February 2020. The worst 1% of days subtract 33% of the total earned by the other 99%. This is the short-volatility payoff profile quantified: steady accumulation punctuated by losses large enough to dominate the average.

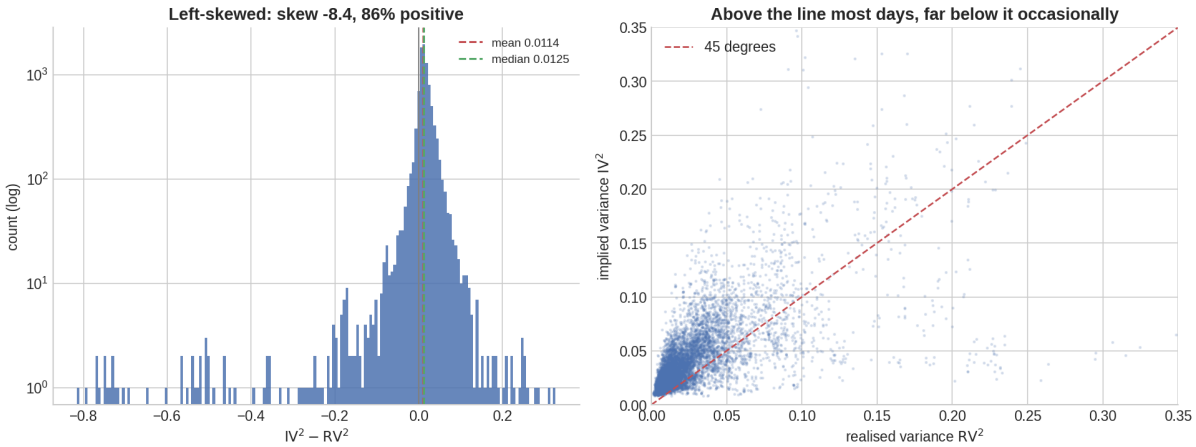



**Figure 7:** Distribution of the premium (log count scale) and the implied-realised scatter. Above the 45-degree line most days, far below it occasionally.

By decade the premium is not uniformly significant: $t_{NW} = 11.18$ (1990–99), 1.97 (2000–09), 5.72 (2010–19), 0.93 (2020–26), with the point estimate more than halving since the 1990s. Two of four subperiods fail at the 5% level. Pooling into a single comfortable number would conceal this.

## 8 Limitations

**Wing extrapolation and truncation.** SVI extrapolates beyond the last quoted strike by construction. For the December density only $1.5 \times 10^{-4}$ of total mass lies outside the quoted range, so the reported moments barely depend on extrapolation; but the density should be read only on the quoted range. One artefact: beyond $K \approx 11{,}200$, well past the last quote at 10,600, the fitted density dips to near zero and recovers because the butterfly constraint binds at the edge of the padded grid. The mass involved is $O(10^{-7})$ and affects no reported figure, but it is named rather than cropped. For the variance swap the truncation exposure is larger: 4.6% of the sum comes from $K/F < 0.70$.

**Overlapping returns.** Both the physical density and the variance risk premium use overlapping windows. For $p$, the nominal 9,192 observations are 156 effective ones; we therefore prefer the filtered estimator, which does not rely on overlapping samples at all. For the premium, HAC inference inflates the standard error 3.9×. Silverman's bandwidth rule, used for the naive KDE, assumes independence and undersmooths badly here; we report the estimator without an ad hoc correction rather than choose an arbitrary one.

**Functional-form error in the wing.** Raw SVI has asymptotically linear total variance; the observed SPX left wing is concave. Fit error reaches 77.7 bp for $k < -0.5$ against 6.6 bp in the core. A more flexible parameterisation, or a constrained spline, would reduce this; five parameters cannot.

**One cross-section, one maturity.** All cross-sectional results use a single snapshot and, for the kernel, a single expiry. There is no term structure of the kernel here and no time-series variation in $q$. The jump correction's extrapolation to the sample mean is correspondingly heroic.

**The projection is not the kernel.** By Proposition 2.1 we recover $\mathbb{E}^{\mathbb{P}}[\xi \mid S_T]$, not $\xi$. Any component orthogonal to the terminal index level is invisible, so non-monotonicity may reflect an omitted state variable rather than a violation of theory.

**Identification of the drift.** The dominant limitation. As Table 3 shows, the qualitative conclusion about monotonicity is determined by an input estimated with a 2.97% annual standard error. No amount of care on the option side repairs this.

# 9 Conclusion

We estimated the projected Radon–Nikodym derivative from SPX options, validating every step against a known answer first. The Breeden–Litzenberger estimator is consistent at order $h^2$ with a leading constant we verify to $4 \times 10^{-5}$, and ill-posed with a noise amplification we derive and then confirm on live quotes, where the naive extraction is 47 times more noise than signal. Arbitrage-constrained SVI removes the problem, yielding a density that integrates to 0.99997 and reprices the forward to $7 \times 10^{-5}$.

The estimated pricing kernel is non-monotone at the point estimate. We do not claim to have found the pricing-kernel puzzle, because the same data with a drift two standard errors higher produces a monotone decreasing kernel. The honest statement is that the shape is not identified at this sample size.

On the variance side, reconstructing the VIX to 0.07% from our own chain validates the log-contract derivation end to end. The premium is positive and significant under HAC inference, but 47% of its measured magnitude is plausibly a jump correction rather than variance risk compensation, its $t$-statistic falls from 21.3 to 5.5 once overlap is handled, and two of four decades fail at conventional levels.

**Reproducibility.** All figures and tables are produced by five numbered scripts run in sequence; no result in this paper is hand-copied.

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